

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE CODE & NAME : CSA16 Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO4: Examine the applicability of clustering algorithms in real-world scenarios, presenting findings through clear reports with effective visualizations.

Assessment Tool Weightage: 40%

QUESTIONS: 05

Total Marks:50

Q1. Dataset: Hospital Patient Segmentation

Patient ID	Age	Monthly Medical Expense (₹)	Health Risk Score
P1	28	12000	65
P2	55	25000	40
P3	42	20000	75
P4	30	11000	70
P5	60	28000	35

SSE Values

K	SSE
1	600
2	350
3	220
4	200
5	190

Question (BL4 – 8 Marks):

A healthcare organization uses K-Means clustering to group patients based on their medical expenses and health risk scores. Interpret the elbow point from the SSE values provided and determine the optimal number of clusters. Discuss how selecting this number of clusters affects patient segmentation and healthcare decision-making.

Q2. Dataset: E-Commerce Order Analysis

Order ID	Product Category	Order Value (₹)
O1	Electronics	55000
O2	Clothing	25000
O3	Electronics	57000
O4	Furniture	30000
O5	Clothing	27000

Question (BL4 – 8 Marks):

An online retailer applies hierarchical clustering to group customer orders based on product category and order value. A dendrogram is generated from the dataset. Interpret how the clusters are formed and identify the appropriate number of clusters. Explain the significance of the selected clusters in understanding customer purchasing behaviour.

Q3. Dataset: Mobile App User Behaviour

User	Daily Usage Time (min)	Login Frequency	In-App Purchase (₹)
U1	120	18	2000
U2	60	5	5000
U3	140	22	1500
U4	50	4	6000
U5	130	20	1800

Question (BL5 – 8 Marks):

A mobile application company uses the HAC algorithm to segment users based on their activity patterns. Interpret how users are assigned to clusters using probability-based membership values. Discuss the advantages of soft clustering over hard clustering for customer behavior analysis.

Q4. Dataset: Agricultural Crop Monitoring

Farm	Rainfall (mm)	Soil Fertility Score	Crop Yield (tons)
F1	300	8	85
F2	500	4	70
F3	400	6	80
F4	280	9	90
F5	520	3	65

Question (BL4 – 8 Marks):

An agricultural research center performs clustering on farm data before and after applying normalization. Interpret the differences observed in the clustering results. Explain how normalization influences cluster formation and improves the reliability of agricultural data analysis.

Q5. Dataset: Employee Performance Evaluation

Employee	Productivity Score	Attendance (%)	Skill Assessment Score
E1	88	92	84
E2	62	75	68
E3	78	88	72
E4	55	65	58
E5	93	96	90

Silhouette Scores

Model	Score
K=2	0.48
K=3	0.68
K=4	0.52

Question (BL5 – 8 Marks):

A company clusters employees based on productivity, attendance, and skill assessment scores. Analyse the silhouette scores obtained for different clustering models and determine the most suitable number of clusters. Justify your answer and explain how the selected model contributes to effective employee performance evaluation.

Course Outcome: CO4 – Examine the applicability of clustering algorithms in real-world scenarios, presenting findings through clear reports with effective visualizations.

The following answers are prepared from the uploaded assessment questions and datasets.

Q1. Hospital Patient Segmentation Using K-Means

Introduction

K-Means clustering is an unsupervised machine learning technique used to divide data into a specified number of groups called clusters. In this problem, a healthcare organization wants to group patients based on monthly medical expenses and health risk scores. The objective is to identify meaningful groups of patients so that healthcare resources can be allocated effectively. The dataset contains five patients with different medical expenses and health risk scores.

Given SSE Values

K	SSE
1	600
2	350
3	220
4	200
5	190

SSE, or Sum of Squared Errors, measures how close the data points are to the centers of their clusters. A smaller SSE indicates more compact clusters.

Elbow Method

The elbow method is used to determine the optimal value of K. The reduction in SSE is:

- $K = 1$ to $K = 2$: $600 - 350 = 250$
- $K = 2$ to $K = 3$: $350 - 220 = 130$
- $K = 3$ to $K = 4$: $220 - 200 = 20$
- $K = 4$ to $K = 5$: $200 - 190 = 10$

The SSE decreases significantly up to $K = 3$. After $K = 3$, the reduction becomes very small. Therefore, the curve shows a clear elbow at $K = 3$.

Optimal Number of Clusters

The optimal number of clusters is $K = 3$. This means that patients can be divided into three groups based on their medical expenses and health risk scores.

- High-risk / high-expense patients
- Medium-risk / medium-expense patients

- Low-risk / low-expense patients

The exact cluster membership would normally be obtained by applying the K-Means algorithm to the patient data.

Effect on Patient Segmentation

Selecting three clusters allows the healthcare organization to distinguish patients according to their healthcare requirements. Patients with higher medical expenses and higher risk scores may require closer monitoring and greater healthcare resources. Patients in a moderate cluster may require regular medical attention, while lower-risk patients may be managed using routine healthcare services.

Importance in Healthcare Decision-Making

- Identify high-risk patient groups.
- Allocate medical resources efficiently.
- Prioritize patients requiring greater attention.
- Understand patterns in medical expenditure.
- Develop targeted healthcare programs.
- Improve patient monitoring and management.

If $K = 5$ were selected simply because it has the lowest SSE, the organization could create unnecessarily many groups. The improvement from $K = 4$ to $K = 5$ is only 10, so the additional complexity is not justified.

Conclusion

The elbow point occurs at $K = 3$, because SSE decreases substantially up to three clusters and then decreases only slightly. Therefore, three clusters are the most appropriate choice for patient segmentation. This provides a balance between cluster quality and simplicity and can support better healthcare planning and decision-making.

Q2. E-Commerce Order Analysis Using Hierarchical Clustering

Introduction

Hierarchical clustering is an unsupervised clustering technique that creates a hierarchy of groups. The grouping process can be represented using a dendrogram. In the given e-commerce problem, customer orders are grouped based on product category and order value. The purpose is to identify similar purchasing patterns.

Given Dataset

Order ID	Product Category	Order Value
O1	Electronics	₹55,000
O2	Clothing	₹25,000
O3	Electronics	₹57,000
O4	Furniture	₹30,000
O5	Clothing	₹27,000

Formation of Clusters

Cluster 1 – Electronics: O1 and O3 both belong to Electronics, and their order values are very close (₹55,000 and ₹57,000). Therefore, they are highly similar and can form one cluster: Cluster 1 = {O1, O3}.

Cluster 2 – Clothing: O2 and O5 both belong to Clothing, and their order values are close (₹25,000 and ₹27,000). Therefore, they can form another cluster: Cluster 2 = {O2, O5}.

Cluster 3 – Furniture: O4 belongs to Furniture and has an order value of ₹30,000. Since it represents a different product category, it can initially remain as a separate cluster: Cluster 3 = {O4}.

Appropriate Number of Clusters

Based on the available dataset, a meaningful interpretation is three clusters.

Cluster	Orders	Interpretation
1	O1, O3	High-value Electronics
2	O2, O5	Clothing orders
3	O4	Furniture order

Role of the Dendrogram

The dendrogram shows the sequence in which orders are merged. Orders that are more similar are joined at lower levels, while dissimilar orders are joined at higher levels. By selecting an appropriate distance level on the dendrogram, the retailer can decide how many clusters should be retained.

Significance for Customer Behaviour

The clusters can help the retailer understand purchasing patterns. The Electronics group has high order values, indicating a high-value product segment. The Clothing group has relatively lower order values and similar purchasing behaviour. Furniture represents another product category that may have different purchasing characteristics.

- Design category-specific marketing strategies.
- Identify high-value orders.
- Recommend related products.
- Understand customer purchasing patterns.
- Plan inventory.
- Develop targeted promotional campaigns.

Conclusion

Hierarchical clustering provides a visual and useful way to identify similarities among e-commerce orders. Based on the given data, three meaningful groups can be identified: Electronics, Clothing, and Furniture. These clusters can help the retailer understand purchasing behaviour and make better marketing and business decisions.

Q3. Mobile App User Behaviour Using Soft Clustering

Introduction

The mobile application company wants to segment users based on their daily usage time, login frequency, and in-app purchase amount. The question focuses on using probability-based membership values to assign users to clusters. Unlike hard clustering, soft clustering allows a user to have different degrees of membership in different clusters.

Given Dataset

User	Daily Usage	Login Frequency	In-App Purchase
U1	120 min	18	₹2,000
U2	60 min	5	₹5,000
U3	140 min	22	₹1,500
U4	50 min	4	₹6,000
U5	130 min	20	₹1,800

Understanding Soft Clustering

In hard clustering, each user belongs to exactly one cluster. In soft clustering, U1, for example, may have 0.85 membership in Cluster A and 0.15 membership in Cluster B. This provides more information about user behaviour.

Possible User Segments

Cluster A – Highly Active Users: These users have high daily usage, high login frequency, and relatively lower purchase values. Examples include U1, U3 and U5.

Cluster B – Lower-Activity, Higher-Purchase Users: These users have lower daily usage, lower login frequency, and higher purchase values. Examples include U2 and U4.

Example Membership Values

User	Active User	High-Purchase User
U1	0.85	0.15
U2	0.25	0.75
U3	0.95	0.05
U4	0.10	0.90
U5	0.90	0.10

These values are illustrative membership values showing the degree of membership rather than a strict yes/no classification. For example, U2 has stronger membership in the high-purchase group, whereas U3 has very strong membership in the active-user group.

Advantages of Soft Clustering

- Handles overlapping behaviour: users can show characteristics of multiple groups.
- Represents uncertainty: membership values show how strongly a user belongs to a particular group.
- Provides better customer segmentation for complex behaviour.

- Supports personalized marketing based on behavioural membership.
- Identifies borderline users for further analysis.

Hard vs Soft Clustering

Hard Clustering	Soft Clustering
User belongs to one cluster	User can have membership in multiple clusters
Simple classification	More flexible classification
Does not show uncertainty	Shows degree of membership
Suitable for clearly separated groups	Suitable for overlapping behaviour

Conclusion

Soft clustering is useful for mobile application user analysis because customer behaviour is not always completely separated into fixed categories. Membership values allow the company to understand the degree to which users belong to different behavioural groups. Therefore, soft clustering can provide more detailed and flexible customer segmentation than hard clustering.

Q4. Agricultural Crop Monitoring and Normalization

Introduction

The agricultural research center performs clustering using farm-related data such as rainfall, soil fertility, and crop yield. The clustering is performed before and after normalization to determine how scaling affects cluster formation.

Given Dataset

Farm	Rainfall (mm)	Soil Fertility	Crop Yield (tons)
F1	300	8	85
F2	500	4	70
F3	400	6	80
F4	280	9	90
F5	520	3	65

Problem Before Normalization

The three variables have very different numerical ranges. Rainfall is approximately 280–520, soil fertility is 3–9, and crop yield is 65–90. Because clustering algorithms commonly use distance calculations, a variable with larger numerical values can have a greater influence on the calculated distance. Therefore, rainfall may dominate the clustering process before normalization.

Example

For F1, rainfall is 300, fertility is 8, and yield is 85. Compared with the fertility score, rainfall has much larger numerical values. Consequently, the clustering algorithm may give greater importance to rainfall even though soil fertility and crop yield are also important agricultural characteristics.

Normalization

Normalization converts different features to a comparable scale. For example, Min-Max normalization can be represented as:

$$X' = (X - X_{\min}) / (X_{\max} - X_{\min})$$

This converts values approximately into the range 0 to 1. After normalization, rainfall, soil fertility, and crop yield can contribute more equally to distance calculations.

Effect on Clustering

Before normalization, the clusters may be influenced strongly by rainfall because its values are much larger. After normalization, all three features have comparable scales. Therefore, the clustering algorithm can consider rainfall, soil fertility, and crop yield more fairly.

Benefits of Normalization

- Prevents feature domination.
- Improves distance calculation.
- Produces more meaningful clusters.
- Improves reliability of agricultural analysis.
- Supports better agricultural decisions.

Agricultural Interpretation

For example, F4 has rainfall of 280 mm, soil fertility of 9, and crop yield of 90 tons. It has relatively lower rainfall but very high fertility and yield. Normalization helps ensure that its overall agricultural characteristics are considered rather than allowing rainfall alone to determine its cluster.

Conclusion

Normalization is important when clustering agricultural data containing variables with different scales. It prevents rainfall from dominating the clustering process and allows soil fertility and crop yield to contribute appropriately. Therefore, normalization improves the reliability and interpretability of agricultural clustering results.

Q5. Employee Performance Evaluation Using Silhouette Score

Introduction

The company uses clustering to group employees based on productivity, attendance, and skill assessment scores. The quality of different clustering models is evaluated using the silhouette score.

Employee Dataset

Employee	Productivity	Attendance	Skill
E1	88	92	84
E2	62	75	68
E3	78	88	72
E4	55	65	58
E5	93	96	90

Given Silhouette Scores

Number of Clusters	Silhouette Score
K = 2	0.48
K = 3	0.68
K = 4	0.52

Meaning of Silhouette Score

The silhouette score evaluates how appropriately each data point belongs to its assigned cluster. A higher score generally indicates better cluster cohesion and better separation between clusters. Therefore, the model with the highest silhouette score should generally be preferred among the given alternatives.

Model Comparison

K = 2 has a silhouette score of 0.48, indicating a moderate level of cluster separation. K = 3 has a score of 0.68, which is the highest among the three models. K = 4 has a score of 0.52, which is higher than K = 2 but lower than K = 3.

Best Model

K = 3, because $0.68 > 0.52 > 0.48$

Possible Employee Segmentation

Cluster 1 – High Performance: employees with high productivity, attendance, and skill scores, such as E1 and E5.

Cluster 2 – Medium Performance: employees with moderate productivity and skill scores, such as E3.

Cluster 3 – Lower Performance: employees requiring greater improvement or support, such as E2 and E4.

The exact cluster assignments would depend on the actual clustering algorithm and distance calculations; the dataset itself does not provide those computed assignments.

Benefits to the Company

- Identify high-performing employees.
- Identify employees requiring additional support.
- Design appropriate training programs.
- Recognize performance patterns.
- Improve workforce planning.
- Develop targeted employee-development strategies.
- Compare productivity, attendance, and skills together.

Why K = 3 is Preferred

K = 3 has the highest silhouette score of 0.68. This indicates that, among the tested models, it gives the most suitable separation and cohesion of employee groups.

Conclusion

The most suitable clustering model is $K = 3$, with a silhouette score of 0.68. Since this is the highest score among $K = 2$, $K = 3$, and $K = 4$, it provides the best clustering structure for employee performance evaluation. The selected model can help the company understand employee performance groups and plan suitable training, recognition, and development activities.